

The Agent Complexity Audit Worksheet

5 Questions to Ask Before Building Any AI Agent

Elliot Betancourt

Most AI agent failures aren't model failures — they're scope failures. Run through these five questions before you write a line of code. Your score tells you how much systems engineering you actually need.

1 Task Boundedness — Can you define start and end?

5 = clear trigger, fixed output, deterministic completion 3 = generally predictable, occasional ambiguity 1 = ad-hoc, human decides when it's done

Agents need clear entry and exit conditions. Fuzzy boundaries mean you're approximating, not automating.

2 Failure Cost — What happens when the model is wrong?

5 = low cost, reversible, human reviews anyway 3 = moderate impact, recoverable with a fix 1 = high cost, cascades downstream, hard to reverse

Your error tolerance determines how much validation and guardrail infrastructure you need.

3 Input Quality — How clean is the incoming data?

5 = structured, consistent, machine-generated 3 = mostly clean, occasional variance 1 = messy, unstructured, user-generated

Clean inputs, simpler prompts. Messy inputs require a preprocessing layer.

4 Output Criticality — Where does the output go?

5 = human reviews before action, low-stakes 3 = semi-automated, some human oversight 1 = directly to production or users, no review

The more direct the path to impact, the more validation sits between model and output.

5 State Requirements — Does this task need memory?

5 = stateless — each call is fully independent 3 = short context window is sufficient 1 = multi-session state required

Stateful agents need explicit state management. This is the most underestimated engineering cost.

Decision Matrix

Total Score	Recommendation
21-25	Trust the model. Minimal scaffolding.
15-20	Hybrid. Add validation and retry logic.
10-14	Agent-assisted. Human in the loop.
Below 10	Systems-level required. Full orchestration.

The model is the easy part. The hard part is knowing when to trust it alone.